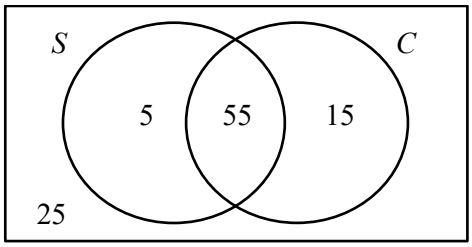
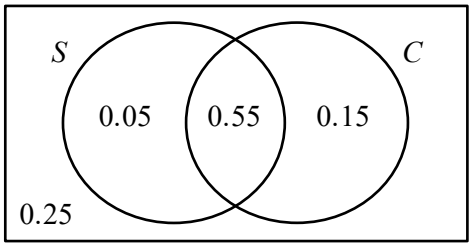
Question	Answer	Marks	Partial Marks
12(a)	 or 	**3**	**If values placed:** **B2** for 55 correctly placed OR **B1** for 25 correctly placed **M1** for total $S = 60$ and total $C = 70$ with intersection not equal zero/not blank OR **Alt method if probabilities placed:** **B2** for 0.55 oe correctly placed OR **B1** for 0.25 oe correctly placed **M1** for total $S = 0.6$ and total $C = 0.7$ with intersection not equal zero/not blank
12(b)(i)	$\frac{70}{100}$ oe	**1**	
12(b)(ii)	$\frac{55}{100}$ oe	**1**	FT $\frac{\textit{their } 55}{100}$ from *their* Venn diagram
12(b)(iii)	$\frac{85}{100}$ oe	**1**	FT $\frac{\textit{their } 25 + \textit{their } 5 + \textit{their } 55}{100}$ or $1 - \frac{\textit{their } 15}{100}$ from *their* Venn diagram
13(a)	$2^8 \times 3^3 \times 5^2 \times 7^{[1]}$ cao	**2**	**M1** for [14 =] 2×7 soi
13(b)	50 or –86 400	**2**	**M1** for $2^3 \times 3^3$ [$\times 5^0$] or $2^6 \times 3^3$ [$\times 5^0$] oe or $2^4 \times 5^2$ or 2×5^2 oe or **SC1** for answers –50, [–]400, [–]1350, [–]3200, [–]10 800, 86 400 oe
14(a)	81	**2**	**B1** for 3^4 or $3^2 \times 9$ or 9×9
14(b)	[±]$\frac{1}{64}$	**2**	**M1** for $\frac{1}{4^3}$ or 64^{-1} or $\frac{1}{\sqrt{4096}}$